

SENSOR BASED SIMULATION USING TINKERCAD/PROTEUS

Automatic Street Light using LDR Sensor

Aim

To simulate an IoT-based automatic Street Light using LDR Sensor in Tinkercad/Proteus where the Street Light turns ON in darkness and OFF in light.

Objective

1. To understand sensor-based automation.
2. To learn Arduino interfacing with sensors.
3. To simulate IoT systems using Tinkercad/Proteus.
4. To control the Street Light using LDR Sensor.

Components Required

<u>S.No</u>	Component	Quantity
1.	Arduino Uno	1
2.	Breadboard	1
3.	LDR Sensor	1
4.	LED	1
5.	Resistors (10k ohm)	2
6.	Jumper wires	Few

Software Required

- Tinkercad
- Proteus Design Suite
- Arduino IDE

Working Principle

An LDR changes its resistance according to the amount of light falling on it:

- Daytime: High light intensity → Low LDR resistance → Street light OFF.
- Nighttime: Low light intensity → High LDR resistance → Street light ON.

Circuit Connections

- LDR connected to Analog Pin A0.
- LED connected to Digital Pin 13 through a resistor.
- LDR forms a voltage divider with a resistor.
- Arduino powered through USB or external supply.

Arduino Program

```
// Automatic Street Light using LDR
```

```
int ldrPin = A0;    // LDR connected to Analog Pin A0
```

```
int ledPin = 13;    // LED connected to Digital Pin 13
```

```
int ldrValue = 0;    // Variable to store LDR value
```

```
int threshold = 500; // Adjust based on your environment
```

```
void setup()
```

```

{
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  ldrValue = analogRead(ldrPin);

  Serial.print("LDR Value: ");
  Serial.println(ldrValue);

  if (ldrValue < threshold)
  {
    // Dark condition
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    // Bright condition
    digitalWrite(ledPin, LOW);
  }

  delay(100);
}

```

Code Explanation

1.Variable Declaration

```

C++
int ldrPin = A0;
int ledPin = 13;

```

`int ldrValue = 0;`

`int threshold = 500;`

`ldrPin = A0` → The LDR sensor is connected to analog pin A0.

`ledPin = 13` → The LED (street light) is connected to digital pin 13.

`ldrValue` → Stores the value read from the LDR.

`threshold` → A reference value used to determine whether it is day or night.

2.Setup Function

C++

`void setup()`

`{`

`pinMode(ledPin, OUTPUT);`

`Serial.begin(9600);`

`}`

This function runs only once when the Arduino starts.

`pinMode(ledPin, OUTPUT);`

Sets the LED pin as an output so the Arduino can control the LED.

`Serial.begin(9600);`

Starts serial communication at 9600 baud rate.

Used to display LDR readings on the Serial Monitor.

3.Loop Function

C++

`void loop()`

`{`

The `loop()` function runs continuously.

4. Reading the LDR value

C++

```
ldrValue = analogRead(ldrPin);
```

Reads the voltage from the LDR circuit.

Returns a value between 0 and 1023.

The value depends on the amount of light falling on the LDR.

5. Displaying the value

C++

```
Serial.print("LDR Value: ");
```

```
Serial.println(ldrValue);
```

Prints the LDR value on the Serial Monitor.

Helps in finding a suitable threshold value.

Example output:

LDR Value: 320

LDR Value: 350

LDR Value: 700

6. Decision Making

C++

```
if (ldrValue < threshold)
```

```
{
```

```
    digitalWrite(ledPin, HIGH);
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(ledPin, LOW);
```

```
}
```

The Arduino compares the LDR value with the threshold.

Condition 1:

C++

`ldrValue < threshold`

Indicates low light (night).

LED turns ON.

Condition 2:

C++

`ldrValue >= threshold`

Indicates bright light (day).

LED turns OFF.

7.Delay

C++

`delay(100);`

Waits for 100 milliseconds before taking the next reading.

Prevents rapid switching and reduces processor load.

Flow of Operation

Plain text

Start

↓

Read LDR Value

↓

Is LDR Value < Threshold?

↓

Yes → Turn LED ON

↓

No → Turn LED OFF

↓

Repeat Continuously

Project Outcome

During the day, sunlight falls on the LDR, so the LED remains OFF.

During the night, light intensity decreases, so the LED turns ON automatically.

This system helps save energy and reduces manual operation of street lights.

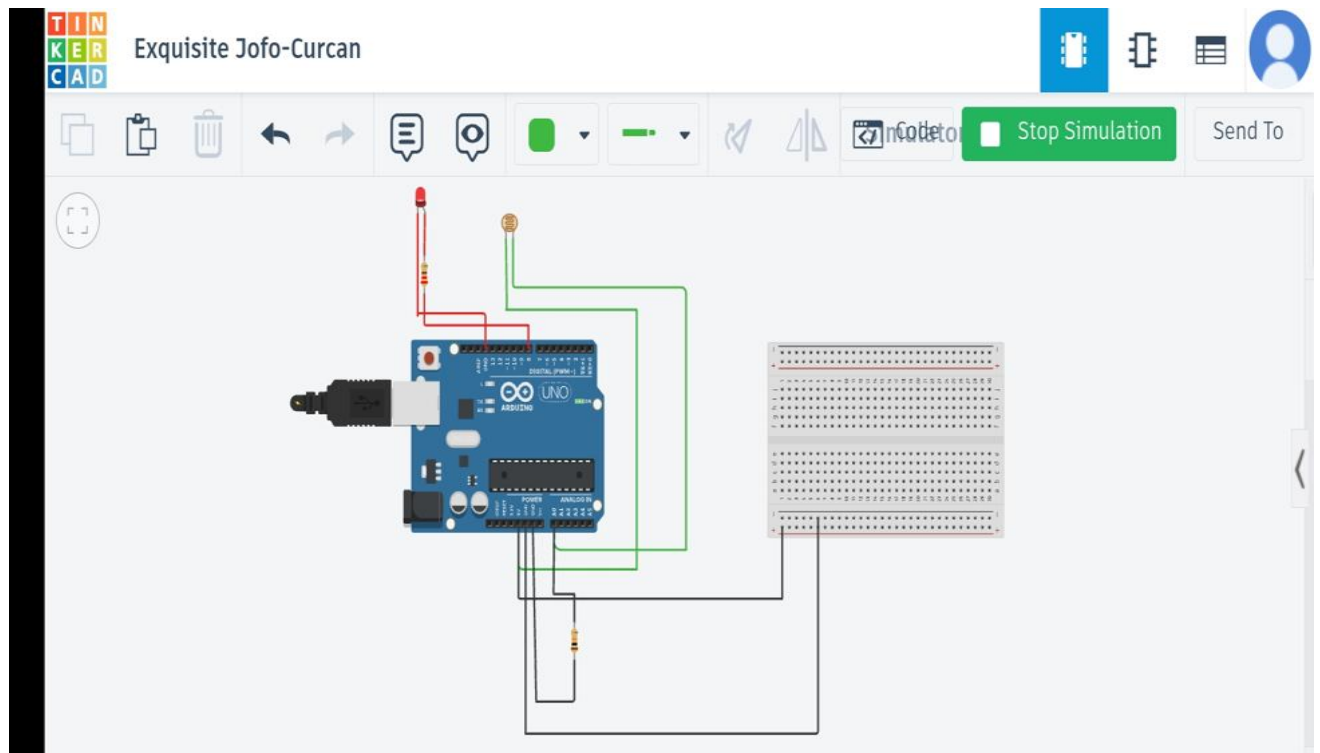
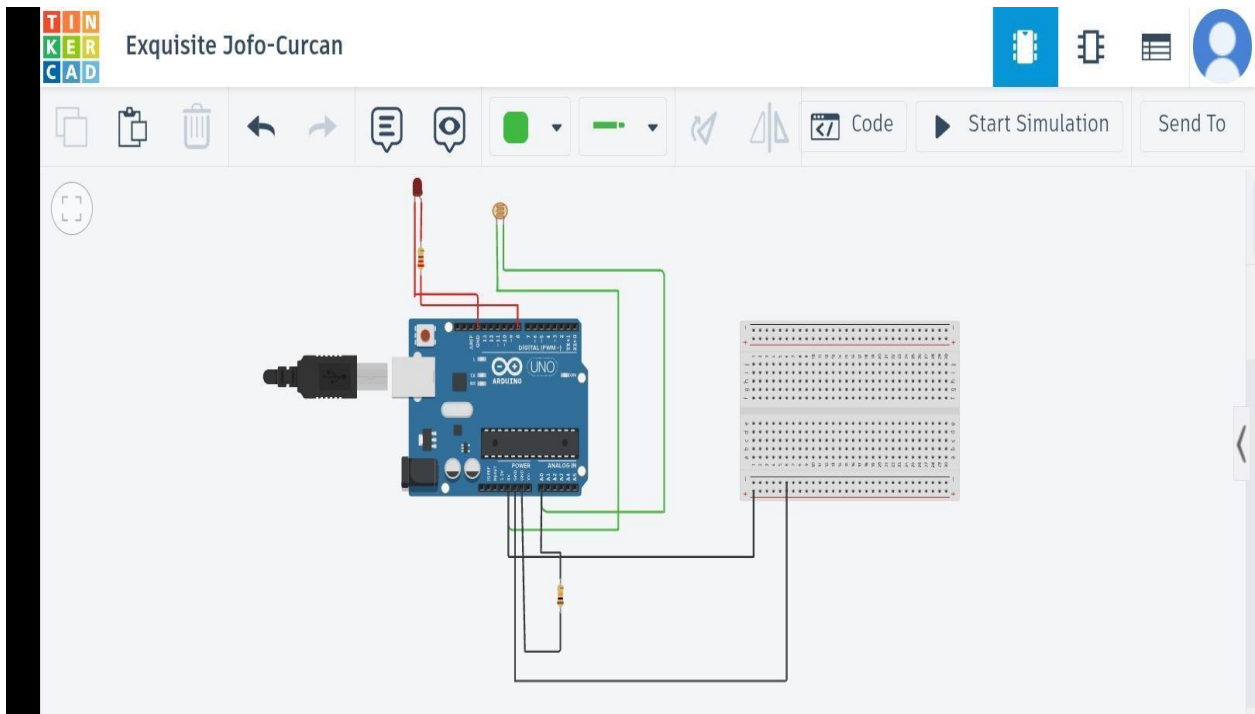
Simulation Steps in Tinkercad

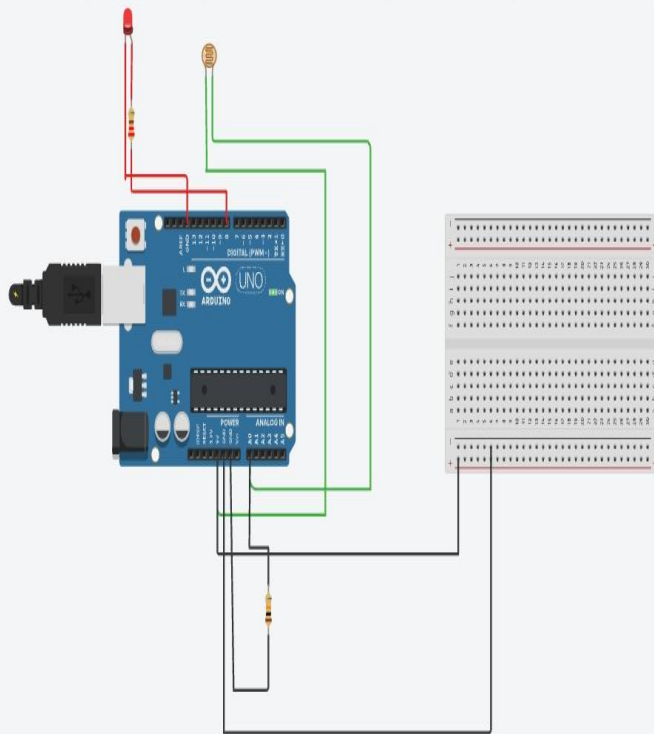
1. Open Tinkercad.
2. Click on "Create New Circuit".
3. Add Arduino Uno.
4. Add LED, LDR, and Resistors.
5. Make the circuit connections.
6. Copy and paste the Arduino code.
7. Click "Start Simulation".
8. Adjust the LDR light intensity.
9. Observe LED turning ON/OFF automatically.

Simulation Steps in Proteus

1. Open Proteus Design Suite.
2. Create a new project.
3. Add Arduino Uno.
4. Add LED and LDR components.
5. Connect all components.
6. Compile Arduino code using Arduino IDE.
7. Generate HEX file.
8. Upload HEX file to Arduino in Proteus.
9. Run the simulation.
10. Observe the LED behaviour.

Expected Output





```
Text 1 (Arduino Uno R3)

1 // Automatic Street Light using LDR
2 int ldrPin = A0;      // LDR connected to Analog Pin A0
3 int ledPin = 13;      // LED connected to Digital Pin 13
4 int ldrValue = 0;     // Variable to store LDR value
5 int threshold = 500;  // Adjust based on your environment
6 void setup(){
7   pinMode(ledPin, OUTPUT);
8   Serial.begin(9600);}
9 void loop(){
10    ldrValue = analogRead(ldrPin);
11    Serial.print("LDR Value: ");
12    Serial.println(ldrValue);
13    if (ldrValue < threshold) {
14      // Dark condition
15      digitalWrite(ledPin, HIGH);
16    }
17    else
18    {
19      // Bright condition
20      digitalWrite(ledPin, LOW);
21    }
22    delay(100);
23  }
24 }
```

Applications

- Street Lighting
- Highways and roads
- Solar street light systems
- Industrial Areas
- Security Lighting
- Smart City projects

Advantages

- Energy saving
- Less human intervention
- Increased safety and security
- Environment-friendly operation
- Reduced electricity bills
- Easy implementation

Conclusion

The Automatic Street Light using an LDR Sensor is an energy-efficient and automated lighting system that switches lights ON during darkness and OFF during daylight. It reduces power consumption, minimizes human intervention, and provides a reliable solution for street lighting. This project demonstrates the practical application of sensors and automation in creating smart and sustainable lighting systems .